Basics of FE-Analysis BK10A6400 Lecture 11

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Consider a single degree-of-freedom mass–spring system:

$$m\ddot{u}(t) + ku(t) = 0 \tag{1}$$

Assume harmonic motion in the form

$$u(t) = \varphi\, \mathrm{e}^{i\omega t} \tag{2}$$

Substitution into the equation of motion gives

$$\left(k - \omega^2 m\right)\varphi = 0 \tag{3}$$

A nontrivial solution requires

$$k - \omega^2 m = 0 \quad \Rightarrow \quad \omega^2 = \frac{k}{m} = \lambda \tag{4}$$

- Eigenvalue: $\lambda = \dfrac{k}{m}$
- Natural angular frequency: $\omega = \sqrt{\dfrac{k}{m}}$
- Eigenmode (scalar): $\varphi \neq 0$ (amplitude can be chosen arbitrarily)

Relation between λ, ω, and f

In eigenvalue analysis of structural dynamics, the quantities λ, ω, and f describe the same physical phenomenon (the natural vibration of the system).

1) Eigenvalue λ

$$(\boldsymbol{K} - \lambda \boldsymbol{M})\,\boldsymbol{\varphi} = \mathbf{0}$$

$$\lambda = \omega^2$$

- Mathematical eigenvalue obtained directly from the generalized eigenvalue problem.
- Unit: $\mathrm{rad}^2/\mathrm{s}^2$
- Not directly a physical vibration quantity, but convenient for numerical solution.

Relation between λ, ω, and f

In eigenvalue analysis of structural dynamics, the quantities λ, ω, and f describe the same physical phenomenon (the natural vibration of the system).

2) Natural angular frequency ω

$$\omega = \sqrt{\lambda} \qquad [\mathrm{rad/s}]$$

- Determines the oscillation rate in the harmonic solution

$$\boldsymbol{u}(t) = \boldsymbol{\varphi}\, \mathrm{e}^{i\omega t}$$

- Widely used in theoretical work and FEM formulations.

Relation between λ, ω, and f

In eigenvalue analysis of structural dynamics, the quantities λ, ω, and f describe the same physical phenomenon (the natural vibration of the system).

3) Natural frequency f

$$f = \frac{\omega}{2\pi} \qquad [\frac{1}{s} \quad \text{i.e} \quad \mathrm{Hz}]$$

- Easy to interpret intuitively (rounds per second)
- Number of cycles per second.
- Commonly used in experiments, vibration testing, and reporting of results.
- Vibration measurement devices usually output both: Hz (f) and rad/s (ω)
- Thus, f is the de facto engineering standard.

Recap: Example: 2DOF mass–spring system

Consider a simple 2DOF system: Two equal masses m are connected with springs k and attached to a fixed wall.

- Generalized displacement vector:

$$\boldsymbol{u}(t) = \begin{Bmatrix} u_1(t) \\ u_2(t) \end{Bmatrix}$$

Mass and stiffness matrices:

$$\boldsymbol{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}, \qquad \boldsymbol{K} = \begin{bmatrix} 2k & -k \\ -k & k \end{bmatrix} \tag{5}$$

Recap: Example: 2DOF mass–spring system

Free undamped vibration:

$$\boldsymbol{M}\ddot{\boldsymbol{u}}(t) + \boldsymbol{K}\boldsymbol{u}(t) = \boldsymbol{0} \tag{6}$$

Assume

$$\boldsymbol{u}(t) = \boldsymbol{\varphi}\,\mathrm{e}^{i\omega t} \tag{7}$$

Substitution gives the generalized eigenvalue problem:

$$(\boldsymbol{K} - \underbrace{\boldsymbol{\omega}^2}_{\boldsymbol{\lambda}}\boldsymbol{M})\boldsymbol{\varphi} = \boldsymbol{0} \tag{8}$$

where $\boldsymbol{\omega}$ is a vector of angular eigenfrequencies and $\boldsymbol{\varphi}$ is a matrix of eigenmodes. Eigenvalues $\lambda_i = \omega_i^2$ and eigenmodes $\boldsymbol{\varphi}_i$ follow from

$$\det(\boldsymbol{K} - \lambda\boldsymbol{M}) = 0$$

Note that MATLAB (and other numerical softwares) gives $\boldsymbol{\lambda}$ as a matrix due to computational reasons.

Recap: Example: 2DOF mass–spring system

In MATLAB, the solution can be found with the commands eig (or eigs).

```
m = 1.0; k = 100.0;
M = [ m 0 ; 0 m ];
K = [ 2*k -k ; -k k ];
[Phi, Lambda] = eig(K, M);
lambda = diag(Lambda); omega = sqrt(lambda); f =
omega/(2*pi);
```

Eigenvalue solvers in MATLAB: `eig` vs. `eigs`

In MATLAB there are two main eigenvalue solvers used in FEM and dynamics: `eig` and `eigs`.

`eig`: dense, all eigenvalues

- Intended for dense or moderately sized matrices.
- Computes *all* eigenvalues (and optionally all eigenvectors).
- Based on direct methods (QR / QZ, Schur decomposition), not ARPACK-type iterative solvers.

Examples

- Standard problem:

$$[V, D] = \mathtt{eig}(A) \quad \Rightarrow \quad AV = VD$$

- Generalized problem:

$$[V, D] = \mathtt{eig}(K, M) \quad \Rightarrow \quad KV = MVD$$

Eigenvalue solvers in MATLAB: `eig` vs. `eigs`

In MATLAB there are two main eigenvalue solvers used in FEM and dynamics: `eig` and `eigs`.

`eigs`: sparse, a few eigenvalues

- Intended for large (typically sparse) matrices.
- Computes only a small number of selected eigenvalues/eigenvectors.
- Uses iterative ARPACK methods (Arnoldi/Lanczos).

Examples

- First n eigenpairs of A:

$$[V, D] = \mathtt{eigs}(A, n)$$

- First n natural frequencies (smallest) from (K, M):

$$[V, D] = \mathtt{eigs}(K, M, n, '\mathtt{smallestreal}')$$

Eigenvalue solvers in Python: dense vs. sparse

Python (NumPy / SciPy) provides analogous functions to MATLAB's `eig` and `eigs`.

Dense problems: `numpy.linalg.eig` / `scipy.linalg.eig`

- Similar role as MATLAB's `eig`.
- For dense or moderately sized matrices.

Examples (Python)

- Standard eigenproblem (dense):

```
w, V = np.linalg.eig(A)   A V = V diag(w)
```

- Generalized eigenproblem (dense, SciPy):

```
w, V = eig(K, M)   K V = M V diag(w)
```

Eigenvalue solvers in Python: dense vs. sparse

Python (NumPy / SciPy) provides analogous functions to MATLAB's `eig` and `eigs`.

Sparse / large problems: `scipy.sparse.linalg.eigs` / `eigsh`

- Analogous to MATLAB's `eigs`.
- Uses ARPACK (iterative Krylov methods).
- Designed for sparse matrices and a few eigenpairs.

Examples (Python)

```
Complex / non-symmetric: w, V = eigs(A, n=5)
5 eigenpairs
Real symmetric/Hermitian: w, V = eigsh(K, n=5,
M=M, which='SM')
```

Principle of virtual work of the equations of motion (Variational form)

$$
\begin{aligned}
&\int_V \rho \frac{\partial^2 \boldsymbol{u}_h}{\partial t^2} \cdot \delta \boldsymbol{u}_h \, dV + \int_V \boldsymbol{c} \cdot \frac{\partial \boldsymbol{u}_h}{\partial t} \cdot \delta \boldsymbol{u}_h \, dV + \int_V \boldsymbol{\sigma}^T \delta \boldsymbol{\varepsilon}, \mathrm{d}V \\
&= \int_V \boldsymbol{b}^T \delta \boldsymbol{u}_h \, \mathrm{d}V + \int_\Gamma \boldsymbol{t}^T \delta \boldsymbol{u}_h \, \mathrm{d}\Gamma + \sum \boldsymbol{f}_i \delta \boldsymbol{u}_i,
\end{aligned}
\tag{9}
$$

Linearized dynamics

Discretized form of equilibrium (balance law for linear momentum):

$$\boldsymbol{M}\ddot{\boldsymbol{u}} + \boldsymbol{C}\dot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}_{ext}$$

(10)

where $\boldsymbol{C}$ is damping matrix and $\boldsymbol{M}$ is a mass matrix. Consistent mass matrix can be written as follows:

$$\boldsymbol{M} = \int_V \rho \boldsymbol{N}^T \boldsymbol{N} \, \mathrm{d}V = \int_V \rho \boldsymbol{N}^T \boldsymbol{N} \, \det(\boldsymbol{J}_e) \mathrm{d}\boldsymbol{\xi} \tag{11}$$

Rayleigh'n damping is often used in FE packages where damping matrix is assumed to be linear combination of mass and stiffness matrices as follows

$$\boldsymbol{C} = \alpha \boldsymbol{M} + \beta \boldsymbol{K} \tag{12}$$

where α and β are defined experimentally.

Mass matrix

- Consistent mass matrix
  - Formulated using same shape functions with mass matrix
  - has also nondiagonal terms
- Lumped mass matrix
  - Only diagonal terms
  - Computationally efficient
  - Less accurate than consistent mass matrix.
- Hybrid mass matrix
- Combination of previous ones
- Consistent and lumped mass matrices are mostly used.

Lumped mass matrix

To derive the lumped mass matrix, you could think mass is "lumped" at the nodes. The lumped mass matrix should fulfill the requirement of mass preservation as follows:

$$\sum_i M_{ii} = \rho \int_V \mathrm{d}V \tag{13}$$

$$= \int_{-1}^{1} \int_{-1}^{1} \int_{-1}^{1} \det \boldsymbol{J}_e \, \mathrm{d}\xi \mathrm{d}\eta \mathrm{d}\zeta \tag{14}$$

where M_{ii} are (diagonal) components of the lumped mass matrix. Note that the lumped mass matrix has only diagonal values and therefore, it gives some computational savings in linear algebra.

Consistent mass matrix

To derive the consistent mass matrix, you need to write kinetic energy:

$$
\begin{aligned}
W_{kin} &= \frac{1}{2}\rho \int_V \dot{\boldsymbol{u}}_h^T \dot{\boldsymbol{u}}_h \, \mathrm{d}V && (15)\\
&= \frac{1}{2}\rho \int_V (\boldsymbol{N}_m \dot{\boldsymbol{u}})^T \boldsymbol{N}_m \dot{\boldsymbol{u}} \, \mathrm{d}V && (16)\\
&= \frac{1}{2}\dot{\boldsymbol{u}}^T \underbrace{\rho \int_V \boldsymbol{N}_m^T \boldsymbol{N}_m \, \mathrm{d}V}_{\boldsymbol{M}_{con}} \dot{\boldsymbol{u}} && (17)\\
&= \frac{1}{2}\dot{\boldsymbol{u}}^T \underbrace{\rho \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \boldsymbol{N}_m^T \boldsymbol{N}_m \det(\boldsymbol{J}_e) \, \mathrm{d}\xi \mathrm{d}\eta \mathrm{d}\zeta}_{\boldsymbol{M}_{con}} \dot{\boldsymbol{u}} && (18)
\end{aligned}
$$

where $\boldsymbol{M}_{con}$ is a consistent mass matrix of an element. Note that the consistent mass matrix has also some off-diagonal values.

Eigenvalue analysis

In dynamic tasks, system natural frequencies and modes are often needed to know. These can be determined from the equations of motion when the force vector is set to zero. Most commonly, the structural damping is so small that it does not affect the natural frequencies. When the damping matrix is omitted, the equations of motion for free undamped vibrations can be written as follows

$$\boldsymbol{M}\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{0}$$

(19)

Its general solution can be written in a form

$$\boldsymbol{u}(t) = \boldsymbol{\varphi}\mathrm{e}^{i\omega t} \tag{20}$$

Eigenvalue analysis

where φ is the vibration mode (eigenmode), i is the imaginary unit, ω is the natural angular frequency, and t is time. The characteristic equation is obtained by substituting the trial in Eq.20 into the equation of motion.

$$(\boldsymbol{K} - \omega^2 \boldsymbol{M})\boldsymbol{\varphi} = \mathbf{0} \tag{21}$$

where $\boldsymbol{K}$ is $n \times n$ size of stiffness matrix and $\boldsymbol{M}$ is $n \times n$ size of mass matrix. natural angular frequency $\omega = \sqrt{\lambda}$ (rad/s) can be written with eigenvalue λ. The Eq. (21) has the trivial solution if $\boldsymbol{\varphi} = \mathbf{0}$ and $\det(\boldsymbol{K} - \lambda \boldsymbol{M}) \neq 0$. Nontrivial solutions exist if and only if

$$\det(\boldsymbol{K} - \lambda \boldsymbol{M}) = \mathbf{0} \tag{22}$$

where $\lambda_1 \ldots \lambda_n$ can be solved. In this course, we use MATLAB's eigs function to solve eigenvalues and modes. See more about the bonus task.

Eigenvalue analysis

Natural angular frequency can also be written:

$$\omega = 2\pi f \tag{23}$$

where f is natural eigenfrequency (1/s=Hz). Eigenvalues and eigenmodes are usually solved numerically.

Rod (bar, truss) element: stiffness and mass matrices

For a 2-node axial bar element (length L_e, area A_e, Young's modulus E, density ρ):

- Element DOFs: axial displacements at nodes 1 and 2

$$\overline{\boldsymbol{u}} = \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}$$

Consistent element stiffness matrix:

$$\overline{\boldsymbol{K}} = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{24}$$

Consistent element mass matrix:

$$\overline{\boldsymbol{M}} = \frac{\rho A_e L_e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \tag{25}$$

Rod (bar, truss) element: stiffness and mass matrices

The global matrices $\boldsymbol{K}_G$ and $\boldsymbol{M}_G$ are obtained by assembling all elements. In this simple case (one element, free-free modes), $\boldsymbol{K}_G = \overline{\boldsymbol{K}}, \boldsymbol{M}_G = \overline{\boldsymbol{M}}$ Then, the free vibration is again described by

$$(\boldsymbol{K}_G - \omega^2 \boldsymbol{M}_G)\,\boldsymbol{\varphi} = \boldsymbol{0}$$

Example: 2 rod elements (fixed–free)

Consider a bar of total length L divided into two equal elements: $L_1 = L_2 = L/2$, uniform A, E, and ρ.

- Three axial DOFs: u_1, u_2, u_3
- Node 1 is fixed: $u_1 = 0$

Global stiffness matrix (symbolically):

$$\boldsymbol{K}_G = \frac{2EA}{L}\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \tag{26}$$

Global consistent mass matrix:

$$\boldsymbol{M}_G = \frac{\rho AL}{12}\begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 2 \end{bmatrix} \tag{27}$$

Example: 2 rod elements (fixed–free)

After applying $u_1 = 0$ (fixed end), we obtain a 2×2 generalized eigenvalue problem for $\{u_2, u_3\}^T$:

$$\left(\boldsymbol{K}_{Gred} - \omega^2 \boldsymbol{M}_{Gred}\right) \boldsymbol{\varphi}_r = \boldsymbol{0}$$

The eigenvalues $\lambda_i = \omega_i^2$ approximate the continuous rod natural frequencies.